# LE, Bach

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Google Scholar: 1618 citations, h-index: 15, i10-index: 17
https://scholar.google.com/citations?user=AUJWzE8AAAAJ&hl=en

"Curiosity is the wick in the candle of learning" – William Arthur Ward

## Summary

I am interested in Software Engineering (SE), particularly Empirical Software Engineering (Human/User Studies), AI for SE and SE for AI, and Program Analysis. I built systems that involve mining massive amounts of past data, e.g., code bug fixes, for actionable insights to inform future software development. My research has attracted **~$500K AUD as Sole PI** from the Australian Research Council. I am also a collaborator of more than **$2M AUD** of research funding awarded by NSF (USA), and Singapore Ministry of Education. My work has inspired the first-ever automated bug fixing system, namely GetAFix, deployed at Meta in 2019.

I am currently managing a ~$500K AUD of funding from ARC DECRA as sole PI, supervising three PhD students, have graduated three master's students by research, and led 40 master's students (~10 students per group) working with industry partners to deliver various successful projects such as web design, blockchain-based payment system, etc. Under my primary supervision, my students have won prestigious prizes such as **Distinguished Paper Award**, **Google PhD Fellowship**, **University Endeavor Awards, and Best Undergraduate Thesis Award**.

I lecture on various subjects at The University of Melbourne such as Models of Computation (600 students), Programming Language and Implementation – Compiler Theory (45 students), Object-oriented Software Development (400 students), etc.

## Appointments

**Assistant Professor | The University of Melbourne, Australia | July'19 – Present, officially Tenured/Confirmed from September 2023.**

**ARC DECRA Fellow – Sole PI of $488K AUD awarded by the Australian Research Council for Discovery Early Career Researcher| January 2022 – 2024**

## Education

### PostDoc | Carnegie Mellon University, Silicon Valley, USA | June'18 – June'19

- *Project*: ISSTAC: Integrated Symbolic Execution for Space-Time Analysis of Code, aiming at software vulnerabilities detection/prevention, funded by DARPA for Space-Time Analysis of Cyber-Security.
- *Advisor*: ACM Distinguished Scientist & Associate Prof. Corina Pasareanu

### Ph.D. Residency Training | Carnegie Mellon University, Pittsburgh, USA | Aug'15 – June'16

- *Project*: Automatic software repair, as part of research collaboration program LARC-CMU of Singapore Management University with Carnegie Mellon University as a strategic partner.
- *Advisor*: Associate Prof. Claire Le Goues

**Ph.D. Candidate | Singapore Management University, Singapore | Jan'14 – May'18**

- *Thesis*: "Overfitting in Automated Program Repair: Challenges and Solutions". Automatic software repair using Machine Learning, Data Mining, Program Synthesis, Analysis and Verification.
- *Advisor*: Prof. David Lo

**B. Tech. | Hanoi University of Science and Technology, Vietnam | Aug'07 – May'12**

- *Thesis*: HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic.
- Degree obtained under collaboration program with Keio University, Japan, supported by Japanese ODA
- *Advisors*: Associate Prof. Quyet-Thang Huynh and Associate Prof. Wei-Ngan Chin

## Additional Research Experience

**Research Assistant | National University of Singapore, Singapore | Feb'12 – May'12 & Aug'12 – Nov'13**

- *Project*: HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic, aiming at verifying C-like programs using SAT, Separation Logic, and Hoare-style deductive verification.
- *Advisor*: Associate Prof. Wei-Ngan Chin

**Research Intern | Nagoya University, Japan | Aug'11**

- *Project*: Building a Twitter-like social network using C#.
- *Advisor*: Prof. Toyohide Watanabe

## Industry Internship

**FSoft (FPT Software) | Vietnam | Aug'11**

- *Project*: Building a web service application using JSP, as part of undergraduate internship program collaborated between Hanoi University of Science and Technology, and FPT Software company.
- *Company website*: https://www.fpt-software.com/

## Certificates

- Fundamental Information Technology Engineer Skill Standards (FE) Certification, **Japan IT Engineers Examination Center**. Certified by the Ministry of Science and Technology, 2011

## Honors & Awards

- **Distinguished Paper Award** at The 39th International Conference on Software Maintenance and Evolution (ICSME), 2023.
- PhD Scholarship at Singapore Management University, 2014 – 2018.
- PhD Exchange Scholarship to Carnegie Mellon University, 2015 – 2016.
- Best Undergraduate Thesis, Higher Education Development Support Project on ICT (HEDSPI), Hanoi University of Science and Technology (HUST), 2012.
- Undergraduate Internship Scholarship to Nagoya University, 2011.

- First Prize in Mathematics (best among ~200 selected students), Nghe An Province, Vietnam, 2005.
- First Prize in Mathematics (best among ~200 selected students), Nghe An Province, Vietnam, 2000.

## Languages

- **English**: Professional working level
- **Japanese**: Intermediate level (JLPT Level 2, obtained in 2009)
- **Vietnamese**: Native

## Computer Skills

- **Programming**: Scala, Java, Ocaml, Bash, Python, Haskell, Latex, C, C#, C++, Prolog
- **Machine learning**: Classification (WEKA library), Active Learning (JCLAL library)
- **Databases**: PostgreSQL, MYSQL, Google BigQuery
- **Eclipse Plugin**: See https://xuanbachle.github.io/semanticsrepair/

Note: I developed several projects using the above programming languages during my PhD study. The largest project that I developed involves a mix of 14K lines of code in Scala and 3K lines of code in Java.

## Articles

**Note: My students are indicated by double underline.**

[31] **Are We Ready to Embrace Generative AI for Software Q&A?**
Bowen Xu, Thanh-Dat Nguyen, Thanh Le-Cong, Thong Hoang, Jiakun Liu, Kisub Kim, Chen GONG, Changan Niu, Chenyu Wang, Xuan-Bach D. Le, David Lo
The 38th IEEE/ACM International Conference on Automated Software Engineering **(ASE)**, New Ideas Track, 2023.

[30] **The Future Can't Help Fix The Past: Assessing Program Repair In The Wild.**
Vinay Kabadi, Dezhen Kong, Siyu Xie, G. Arthar, Lingfeng Bao, Tien-Duy BL, Xuan-Bach D. Le, and David Lo
The 39th IEEE International Conference on Software Maintenance and Evolution **(ICSME)**, 2023.
Won **Distinguished Paper Award** – the only paper among the accepted paper that won the prize. (Acceptance rate ~22%)

[29] **Multi-Granularity Detector for Vulnerability Fixes.**
Truong Giang Nguyen, Thanh Le-Cong, Hong Jin Kang, Ratnadira Widyasari, Chengran Yang, Zhipeng Zhao, Bowen Xu, Jiayuan Zhou, Xin Xia, Ahmed Hassan, Xuan-Bach D. Le, and David Lo
Transactions on Software Engineering **(TSE)** Journal, 2023

[28] **Invalidator: Automated Patch Validation via Dynamic Invariant Inference**.
Cong-Thanh Le, Duc-Minh Luong, Xuan-Bach D. Le, David Lo, Nhat-Hoa Tran, and Quyet-Thang Huynh.
Transactions on Software Engineering **(TSE)** Journal, 2023.

[27] **Chronos: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Reports.**
Y. Lyu, Thanh Le-Cong, Hong Jin Kang, R. Widyasari, Z. Zhao, Xuan-Bach D. Le, Ming Li, David Lo
The 45th IEEE/ACM International Conference on Software Engineering (**ICSE**), Research Track, 2023.

[26] **An Efficient Cyclic Entailment Procedure in a Fragment of Separation Logic.**

Quang-Loc Le, Xuan-Bach D. Le
The 26th International Conference on Foundations of Software Science and Computation Structures (**FoSSaCS**), Research Track, 2023.

[25] **VulCurator: A Vulnerability-Fixing Commit Detector.**
Truong Giang Nguyen, Thanh Le-Cong, Hong Jin Kang, Xuan-Bach D. Le, David Lo
The 21st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Demo Track, 2022.

[24] **AutoPruner: Tranformer-based Call Graph Pruning.**
Thanh Le-Cong, Hong Jin Kang, Truong Giang Nguyen, S. Haryono, David Lo, Xuan-Bach D. Le, and Quyet-Thang Huynh
The 21st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Research Track, 2022.

[23] **FFL: Fine grained Fault Localization for Student Programs via Syntactic and Semantic Reasoning.**
Thanh-Dat Nguyen, Thanh Le-Cong, Duc-Minh Luong, Van-Hai Duong, Xuan-Bach D. Le, David Lo and Quyet-Thang Huynh
The 38th IEEE International Conference on Software Maintenance and Evolution (**ICSME**), Research Track, 2022.

[22] **Test Mimicry to Assess the Exploitability of Library.**
Hong Jin Kang, Truong Giang Nguyen, Xuan-Bach D. Le, David Lo, Corina Pasareanu
The 31st International Symposium on Software Testing and Analysis (**ISSTA**), Research Track, 2022.

[21] **Toward the Analysis of Graph Neural Networks.**
Thanh-Dat Nguyen, Thanh Le-Cong, ThanhVu H Nguyen, Xuan-Bach D. Le, Quyet-Thang Huynh
The 44th International Conference on Software Engineering (**ICSE**), NIER Track, 2022.

[20] **REFIXAR: Multi-version Reasoning for Automated Repair of Regression Errors.**
Xuan-Bach D. Le, Quang-Loc Le
*The 32nd International Symposium on Software Reliability Engineering* (**ISSRE**), Research Track, 2021.

[19] **Usability and Aesthetics: Better Together for Automated Repair of Web Pages.**
Cong-Thanh Le, Xuan-Bach D. Le, Quyet-Thang Huynh, Phi Le Nguyen
*The 32nd International Symposium on Software Reliability Engineering* (**ISSRE**), Research Track, 2021.

[18] **More Reliable Test Suites for Dynamic Program Repair by Using Counterexamples.**
Amirfarhad Nilizadeh, Marlon Calvo, Gary T. Leavens, Corina Pasareanu and Xuan-Bach D. Le
*The 32nd International Symposium on Software Reliability Engineering* (**ISSRE**), Research Track, 2021.

[17] **Exploring True Test Overfitting in Dynamic Program Repair using Formal Methods.**
Amirfarhad Nilizadeh, Gary T. Leavens, Xuan-Bach D. Le, Corina Pasareanu, and David R. Cok
*International Conference on Software Testing Verification and Validation* (**ICST**), Research Track, 2021

[16] **Smart Contract Development: Challenges and Opportunities**.
Z. Weiqin, L. David, S. K. Pavneet, Xuan-Bach D. Le, F. Yang, Zhenyu Chen, and Baowen Xu
*Transactions on Software Engineering* ***(TSE)***, 2019

[15] **SAFFRON: Adaptive Grammar-based Fuzzing for Worst-Case Analysis**
Xuan-Bach D. Le, Corina Pasareanu, Rohan Padhye, David Lo, Willem Visser, and Koushik Sen
*Java Pathfinder Workshop collocated with ASE* ***(JPF)***, 2019.

[14] **Symbolic Pathfinder for SV-COMP**
Yannic Noller, Corina Pasareanu, Aymeric Fromherz, Xuan-Bach D. Le, and Willem Visser
*Competition on Software Verification* (**SV-COMP**), 2019, held at TACAS.

[13] **On Reliability of Patch Correctness Assessment**.
Xuan-Bach D. Le, Lingfeng Bao, David Lo, Xia Xin, Shanping Li, and Corina Pasareanu
In *International Conference on Software Engineering (**ICSE**),* Research Track, 2019

[12] **Overfitting in Semantics-based Automated Program Repair.**
Xuan Bach D Le, David Lo, Ferdian Thung, and Claire Le Goues.
In *International Conference on Software Engineering (**ICSE**), Journal First Track, 2018.*
In *Empirical Software Engineering (**EMSE**) Journal*, 2018.

[11] **S3: Syntax- and Semantics-guided Repair Synthesis via Programming by Examples.**
Xuan Bach D Le, David Lo, Duc-Hiep Chu, Claire Le Goues, and Willem Visser.
In *European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (**ESEC/FSE**)*, Research Track, 2017.

[10] **JFix: Semantics-based Repair of Java Programs via Symbolic Pathfinder.**
Xuan Bach D Le, Duc-Hiep Chu, David Lo, Claire Le Goues, and Willem Visser.
In *26th International Symposium on Software Testing and Analysis (**ISSTA**)*, Demo Track, 2017.

[9] **Xsearch: A Domain-specific Cross-language Relevant Question Retrieval Tool.**
Bowen Xu, Xing Zhen-chang, Xia Xin, David Lo, and Xuan-Bach D Le.
In *European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (**ESEC/FSE**)*, Demo Track, 2017.

[8] **Towards Efficient and Effective Automatic Program Repair.**
Xuan-Bach D Le.
In *Proceedings of the 31st International Conference on Automated Software Engineering (**ASE**)*, Doctoral Symposium, 2016.

[7] **Enhancing Automated Program Repair with Deductive Verification**.
Xuan-Bach D Le, Quang Loc Le, David Lo, and Claire Le Goues.
In *32nd International Conference on Software Maintenance and Evolution (**ICSME**)*, ERA Track, 2016.

[6] **Empirical Study on Synthesis Engines for Semantics-based Program Repair.**
Xuan Bach D Le, David Lo, and Claire Le Goues.
In *32nd International Conference on Software Maintenance and Evolution (**ICSME**)*, ERA Track, 2016.

[5] **Recommending Code Changes for Automatic Backporting of Linux Device Drivers.**
Ferdian Thung, Xuan-Bach D Le, David Lo, and Julia Lawall.
In *32nd International Conference on Software Maintenance and Evolution* (***ICSME***), Research Track, 2016.

[4] **History Driven Program Repair.**
Xuan Bach D Le, David Lo, and Claire Le Goues.
In *23rd International Conference on Software Analysis, Evolution, and Reengineering* (***SANER***), Research

Track, 2016.

[3] **Should Fixing These Failures be Delegated to Automated Program Repair**?
Xuan-Bach D Le, Tien-Duy B Le, and David Lo.
In *26th International Symposium on Software Reliability Engineering* (***ISSRE***), Research Track, 2015.

[2] **Synergizing Specification Miners Through Model Fissions and Fusions**.
Tien-Duy B Le, Xuan-Bach D Le, David Lo, and Ivan Beschastnikh.
In *30th International Conference on Automated Software Engineering* (***ASE***), Research Track, 2015.

[1] **Active Semi-supervised Defect Categorization**.
Ferdian Thung, Xuan-Bach D Le, and David Lo.
In *23rd International Conference on Program Comprehension* (***ICPC***), Research Track, 2015.

PhD thesis: **Overfitting in automated program repair: Challenges and Solutions.**
Xuan Bach D Le
PhD Thesis, Singapore Management University, Singapore, May 2018.

## Recent Unpublished Work

[32] **Adversarial Patch Generation for Automatic Program Repair,** Aziz Alhefdhi, Cong-Thanh Le, Khanh-Hoa Dam, under revision to submit to nearest venue.

[32] **Refining ChatGPT-Generated Code: Characterizing and Mitigating Code Quality Issues,** Yue Liu, Cong-Thanh Le, Ratnadira Widyasari, Chakkrit Tantithamthavorn, Li Li, Xuan-Bach D. Le, David Lo, under submission to TSE'23.

[33] **PatchZero: Zero-Shot Automatic Patch Correctness Assessment,** Xin Zhou, Bowen Xu, Kisub Kim, DongGyun Han, Cong-Thanh Le, Junda He, Xuan-Bach D. Le, David Lo.

[34] **Mining Graphs to Attack Code Models,** Thanh-Dat Nguyen, Zhou Zhang, Xuan-Bach D. Le, Patanamon Thongtanunam, David Lo, under submission.

[35] **GNN-Infer: Property Inference for Graph Neural Network,** Thanh-Dat Nguyen, Hieu Nguyen, Cong-Thanh Le, Xuan-Bach D. Le, David Lo, ThanhVu Nguyen, Corina Pasareanu, under submission.

## Student Supervision

**PhD students:**

- Vinay Kabadi, Part-time PhD student at The University of Melbourne. Thesis topic: empirical study on software debugging, January 2020 - now.
  - Award: **Distinguished Paper Award** at ICSME'23.
- Thanh-Dat Nguyen, Full-time PhD student at The University of Melbourne. Thesis topic: Program Synthesis for Automated Information Extraction, April 2021 - now.
- Cong-Thanh Le, Joining PhD program at The University of Melbourne in January 2023. Cong-Thanh has been under my supervision, graduated from Hanoi University of Science and Technology (HUST) in 2019.
  - Awards: **Best Undergraduate Thesis** from HUST in 2019. **Google PhD Fellowship** in 2023.

**Master's students:**

- Ishan Maholtra, master's student from The University of Melbourne, 2019 – 2020 (graduated). Topic: Automated Repair of student programming assignments.

- Chijiang Yang, master's student from The University of Melbourne, 2020-2021 (graduated). Topic: Unified debugging: how far are we?
- Shubham Parth, master's student from The University of Melbourne, 2020-2021 (graduated). Topic: Smart contract fuzzing.

**Undergraduate students:**

- Thanh Le Cong, undergraduate student at Hanoi University of Science and Technology, 2019 – now
  - Award: **Best Undergraduate Thesis Award**. Topic: Automated Program Repair
- Siyu Xie, female undergraduate student from Zhejiang University, China, visited SMU during 2017 - 2018. This resulted in a full research paper on automatic repair, now under submission.
  - Award: **Distinguished Paper Award** at ICSME'23 (together with Vinay Kabadi)
- Hanliang Zhang, undergraduate student from Peking University, China during Google Summer of Code, 2019

## Project Supervision

I regularly supervise groups of ~10 students per group working on their master's projects across organizations, e.g., university and industry. The projects span one semester (12 weeks) or three semesters (1 year). Notable projects, **that won University Endeavor Award**, include:

- **Reusable coffee cup**: build an online management system that allows users to borrow and return coffee cups when buying coffee. (1-year project, 2019 - 2020)
- **Apollo Music**: build a blockchain-based system that allows the music industry to manage payments for music shows, e.g., online blockchain-based smart contract to pay artists and promoters for music shows. (1-year project, 2020 - present)

## Teaching

- **Lecturer/Coordinator**: Models of Computation, The University of Melbourne, 2022 (~600 students enrolled). This is a core (required) subject in Computer Science at Unimelb.
- **Lecturer/Coordinator**: Models of Computation, The University of Melbourne, 2021 (~600 students enrolled). This is a core (required) subject in Computer Science at Unimelb.
- **Lecturer**: Programming Language Implementation (Compiler Theory), The University of Melbourne, 2020 (~40 students enrolled) – Graduate Level
- **Lecturer/Coordinator**: Object-Oriented Software Development, The University of Melbourne, 2019, 2021, 2023 (~350 students enrolled)
- **Lecturer/Coordinator**: Masters Software Engineering Project, The University of Melbourne, 2020 (~120 students enrolled) – Graduate Level
- **Guest Lecturer**: Program Analysis and Modeling, Graduate level, SMU, 2018
- **Teaching Assistant**: Software Mining and Analysis, Graduate level, SMU, 2018
- **Teaching Assistant**: Analytics Foundations, Undergraduate level course, SMU, 2017

## Activities

- **Invited talks:**
  - Invited speaker at NASA Robust Software Engineering, California, USA. Topic: Recent Advances on Automated Program Repair.

- Top 4 speakers invited to serve on the expert panel to talk about the Future of Automated Program Repair at Automated Program Repair workshop, collocated with ICSE'23 in Melbourne, Australia.
- Invited speaker at Monash University, 2019 and 2023. Topic: Automated Software Debugging and Repair.

- **Charing role**:
  - Chair of Software Engineering Track for the 11th International Symposium on Information and Communication Technology (SoICT), 2022
  - Chair of Demonstration Track for the IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 2022
  - Chair of Student Research Competition for the 36th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2021
- **Committee Member** for:
  - The 45th International Conference on Software Engineering (ICSE), 2023, Research Track
  - The 44th International Conference on Software Engineering (ICSE), 2022, Software Engineering in Practice Track
  - The 43rd International Conference on Software Engineering (ICSE), 2021, New Ideas Track
  - The 36th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2021. Research Track
  - The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2020, Research Track
  - The 35th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2020. Research Track
  - The 42nd International Conference on Software Engineering (ICSE), 2020, New Ideas Track
  - The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Artifact Evaluation Committee, 2019
  - The 19th IEEE International Conference on Software Quality, Reliability, and Security (QRS), 2019
  - International Workshop on Intelligent Bug Fixing (IBF), 2019
- **Reviewer** for:
  - IEEE Transactions on Network and Service Management (TNSM), 2022
  - Transactions on Software Engineering (TSE), 2019, 2020, 2021, 2022
  - Transactions on Software Engineering and Methodology (TOSEM), 2018, 2019, 2020, 2021
  - Software Testing, Verification and Reliability (STVR), 2018
  - Information and Software Technology (IST), 2018
  - Journal of Computer and Communications (JCC), 2018
  - Journal of Systems and Software (JSS), 2018
  - Empirical Software Engineering Journal (EMSE), 2017, 2021
- **Sub-reviewer** for: *2013*: SAC; *2014*: POPL; *2016*: APSEC, ICECCS, MSR (data track), SATE, SAC; *2017*: IWESEP, ICSME, VL/HCC, RV; *2018*: SCAM, ESEC/FSE (Demo Track)
- **Student volunteer**: ASE 2016, Singapore.
- **Oral conference presentations**:
  - ESEC/FSE 2017, Paderborn, Germany.
  - ASE 2016, Singapore.
  - ISSRE 2015, Gaithersburg, USA.

## Software

- JFix and S3: Semantics-based repair of Java programs (ISSTA'17 and FSE'17), built atop Symbolic PathFinder – a notable symbolic execution engine for Java.
  - https://xuanbachle.github.io/semanticsrepair/
- HDRepair: Search-based program repair (SANER'16) using data mining and genetic programming, built atop of gSpan graph miner and Watchmaker evolutionary framework.
  - https://github.com/xuanbachle/bugfixes

## Grants

- External grants:
  - ARC DECRA 2022 (Sole PI, $488K AUD): "*Practical Automated Software Bug Fixing via Syntactic and Semantic Analyses*", $438K AUD awarded by ARC and $50K AUD awarded by The University of Melbourne.
- Internal grants:
  - Grant for hiring a postdoc position starting from January 2024 to December 2024, awarded by The University of Melbourne in recognition of my DECRA'22.
  - Seed grant: "*Formally Verified Syntax- and Semantics-guided Automatic Proof Repair*", with Christine Rizkallah, $8K AUD, awarded by Computer Science group, The University of Melbourne.
  - Sole PI, $20K AUD, "*Refixar: Practical Automated Repair of Software Regression Errors*", awarded by The University of Melbourne.
- Collaborator of research grants (awarded to collaborators, not The University of Melbourne):
  - Collaborator and assisted in writing part of funding proposal: *Uncovering vulnerabilities in Machine Learning Frameworks via Software Composition Analysis and Directed Grammar-based Fuzzing*, awarded $500K SGD by Singapore government. This project is a collaboration between Singapore Management University (PI), Veracode (security company in Singapore, PI), University of Melbourne (collaborator), and Carnegie Mellon University (collaborator).
  - Assisted in writing part of funding proposal: *Human-Guided hybrid Search* (HUGS), a joint proposal between Carnegie Mellon University, University of California Santa Barbara, University of California Berkeley, and University of Wisconsin Madison, awarded NSF Medium $1M USD. I contributed to writing semantics-based repair using symbolic execution and program synthesis.

## References

Available upon request.